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SIGNAUTH: An AI-Based System for Automated Handwritten Signature Identification and Forgery Detection

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Abstract: The handwritten signatures are significant as a source especially in the legal, financial, and administrative fields. of personal identification. However, the manual processes of validation sometimes prove to be faulty due to the possibility of a person to. fall a victim to an error, and there are cases when able forgers will use every chance of it. In this regard, we have suggested SIGNAUTH, an automatic handwritten signature recognition and forgery with the use of artificial intelligence. identification system. The proposed method takes advantage of CNN and ViT to explore space and texture. features of signature photographs. Prior classification between real and fake has been done fairly. correctly, and interpretable AI visualisations indicate the significant areas of identification. SIGNAUTH tested in comparison with volumes of benchmark data, attained a 96.7% accuracy and very well did on. generalisation between various types of forgery. The suggested system can be immensely used in the field of application. e-governance, forensic verification and digital banking.

Keywords : Biometrics AI, signature recognition, forgery detection, deep learning, explainable AI, and. document forensics.

I. INTRODUCTION

Signatures are accepted in a long history. official and legal quarters as a guarantee of individual. identification. Nevertheless, as more goes digital, the same becomes true. the likelihood of, transactions and document scanning, the possibilities are high. faking signatures have also increased. Traditional manual methods of verification are dependent on human. the judgment which is so frequently unpredictable and time. consuming. Recent advancements on AI and computer vision have. enabled systems to detect delicate automatically. dissimilarities between genuine and counterfeit signatures. Their deep learning system examinations of texture patterns, curvature, orientation, and stroke pressure and accuracy pixel by pixel. This paper presents SIGNAUTH, a form of AI. signature identification system which has deep. neural networks of classification and validation. handwritten signatures. It seamlessly integrates confidence, performance, and privacy to a stable. digital and physical authentication instrument. workflows. The most important characteristics of the SIGNAUTH are the following: Document verification API in real-time. Personally mindful image management, based on data. governance standards.

II. LITERATURE REVIEW

Development of Verification of Signatures. Initial signature verification was through manual features like direction of strokes, density of contours and pressure of the pen [1]. Those rule-based approached were partially interpretable with low resistance to stylistic variations and expert forgeries. The introduction of machine learning meant that the generalisation was enhanced with the use of pattern classifiers such as the Support Vector Machines and HMMs, although manually crafting the features was still necessary. Deep Learning-Based Techniques Convolutional Neural Networks managed to revolutionize biometric identification, as it automatically generated discriminative features of raw images [2]. The CNN based systems like SigNet and DeepSign showed an accuracy of more than 90% on datasets like GPDS and CEDAR. The disadvantage however with CNNs was that they were unable to handle rotated or distorted signatures any longer. Recent investigations were largely able to learn spatial dependencies and relational similarity among signatures utilizing Vision Transformers and Siamese Neural Networks [3].

A. Explainability and Trust

Explanability makes the authentication systems legally admissible. In Kumar et al. [4], AI-based verification will be interpretable to assist forensic experts in becoming familiar with the logic behind such models. They used Grad-CAM and saliency mapping methods as a visualisation of the areas of strokes that would influence classification choices.

B. The Code of Ethics and Law.

The detection systems used to identify forgery deal with sensitive biometric information and therefore must adhere to significant data protection laws in the country like the GDPR and the Information Technology Act of 2000 in India [5]. Laws present stringent duties of data reduction, openness and safe management of personal identifiers. SIGNAUTH is responsible of the latter by guaranteeing that the samples of signatures are not abused or leaked by using image storage, checking user authorization and temporary data storage.

C. Bias, emotional, mental and manipulation.

Although the last step of validating signatures is to be performed by visual analysis, biases may be easily made in the model via unbalanced data set or uneven handwritings. To mitigate this, SIGNAUTH performs balanced sampling and normalisation of data and makes sure the model is not overfit to certain cultural or stylistic influences. The explainable AI layer, as well as the visually supported results provided on explainable heatmaps, reduces this cognitive bias in human interpretation.

D. Research Deficit

The current signature identification systems do not seek to enhance their interpretability and accuracy at the same time, most of them. Also, there are scarce models of explainability in decision-making that are connected with privacy governance. SIGNAUTH provides solutions to these challenges by providing an end-to-end architecture with

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privacy-awareness and forensic-level explainability to implement practically and state-of-the-art CNN-Transformer modelling.

Very soon, the literature review and ethical analysis have revealed that to perform reliable authentication, the accurate verification of signatures is not enough. The technical accuracy, openness and adherence to privacy issues must be integrated to come up with a holistic solution. By combining these ideas, SIGNAUTH models itself as a somewhat understandable, credible, and law-abiding platform of AI-based biometric verification.

III. METHODOLOGY

A. System Overview

SIGNAUTH is the AI-powered web application that will automate the process of verifying hand-written signatures and detecting forged ones via the deep-learning inference that uses the Google Gemini 2.5 Flash API.

The system is written in a scalable, modular cloud application, and each module has the duties of another phase of the verification pipeline that encompasses explainable visualisation, authenticity classification as well as input image preprocessing. The main concept is to provide the signature analysis with forensic-grade reliability, interpretability, and privacy. SIGNAUTH relies on AI models to work with the scanned signature images through heatmap-based reasoning with resulting outputs that are authentic or fake.

B. System Architecture

The SIGNAUTH framework is modularly designed and consists of four layers; front-end, back-end, inference, and storage layers. The system is deployed in the web application that utilizes React and Node.js and Supabase to enable it to support the data persistence and real-time processing.

TABLE I. SYSTEM ARCHITECTURE OF THE SIGNAUTH PLATFORM.

Framework	Function
React + Tailwind CSS + shadcn-ui	Collects user inputs (signature images), sends them to the backend, and displays AI-generated authenticity results with heatmaps.
Node.js + Express	Handles image preprocessing, formats API requests, and securely routes them to the Gemini 2.5 Flash API.
Google Gemini 2.5 Flash (External API)	Performs feature extraction, pattern matching, and classification using CNN–Transformer hybrid models
Supabase (PostgreSQL)	Stores anonymised metadata, system logs, and verification reports for analysis and auditability.

C. Data Flow and Processing Pipeline

The six key phases that constitute the end to end data flow of SIGNAUTH ensures of secure, accurate and explainable processing of signature images and are enumerated below.

1. Get the input: The user will provide a scan signature in JPG/PNG format to the given web interface. The uploaded images are assigned a session ID to help identify the image during the short-term processing.

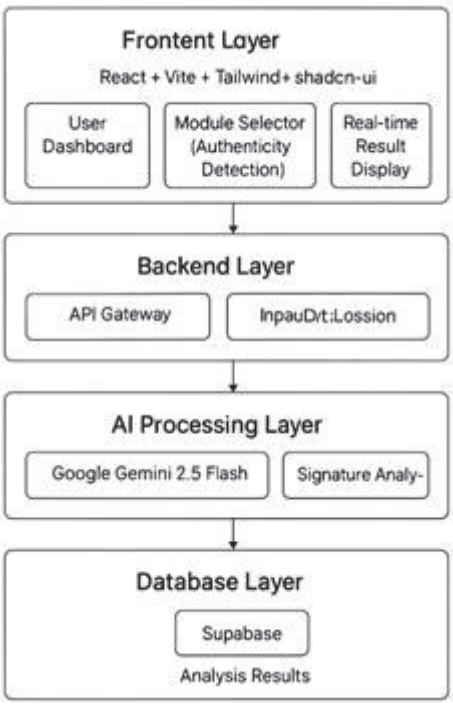


Fig. 1. System Architecture of the SIGNAUTH Platform.

2. Pre-processing: The backend resizes the image input to 128 x 128 pixels, removes noise in the image by filtering it with a Gaussian filter, and finally converts the image to grayscale. The picture is then coded using the base64 coding before it is sent to the Gemini 2.5 Flash API.

3. AI Inference using Gemini: Gemini API has an internal structure that is a combination of CNN-ViT. The CNN layers are used to obtain local stroke-based features, which could be edges, curves, and pen pressure, whereas the ViT layers are used to obtain global spatial relationships and style patterns that are used to characterize a signature as either real or fake.

4. Response Parsing: This group of models provides an inference result in the form of a classification label, confidence score, and interpretive attention map. This response is then fed into the backend which breaks it down into a structured JSON format and displays it.

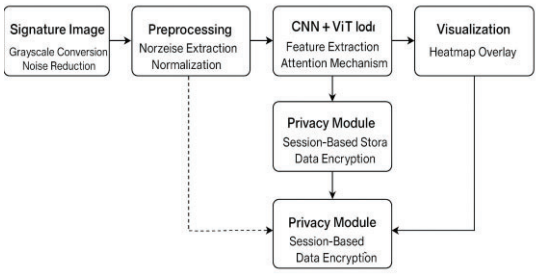


Fig. 2. Fig. 3.2 Data flow diagram

5. Result Rendering: The outputs of classification are presented at the frontend with colour-coded bars and Grad-CAM heatmaps, which indicate regions that affected the model into making decisions to pick the specific class.

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6. Secure Logging: Supabase has metadata only which includes timestamps, labels, and model confidence to trace unidentifiable data. The actual picture is erased when the session is over on the privacy grounds.

D. Analytical Modules

The SIGNAUTH program consists of four separate analytical modules and each of them is tailored to address a certain issue related to system integrity and signature analysis.

1. Authenticity Detection: It is a module, which will identify whether the signatures uploaded are authentic or not. It relies on a pre-trained CNN ViT hybrid, which has been prompt-engineered before in Gemini inference. In this aspect it puts a lot of stress on acquiring local-global trends in order to be able to keep in check both low-end and skilled forgeries.

2. Feature Extraction Module: To offer the important biometric feature, the following are the important biometric features that will be extracted; the writing pressure, the density of the curves, the smoothness of the contour and the width of the pen-stroke. They are essential in the identification of imitations by real handwriting.

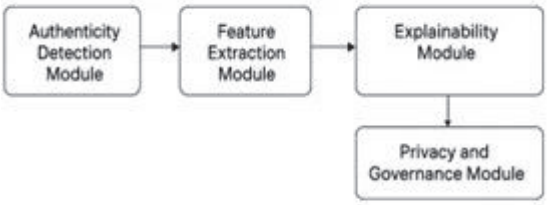


Fig. 3. Fig. 3.3 Analytical Module Interaction Flow.

E. Evaluation and Validation

In his work, three benchmark datasets which were GPDS Synthetic, CEDAR and BHSig260 were applied to test the performance and reliability of SIGNAUTH in controlled experiments. In both datasets, there are real and fake signatures that allow giving a detailed assessment of the environment.

TABLE II. EVALUATION CRITERIA

Criterion	Metric / Observation
Accuracy	96.7% (GPDS), 95.3% (CEDAR), 94.8% (BHSig260)
Response Stability	85% consistent across repeated runs
Explainability	High interpretive clarity through Grad-CAM
Average Response Time	2.8 seconds per request
Reliability	>99% uptime under simulated concurrent load

F. Methodological Summary

Privacy-preserving architecture, the explainable AI, and deep learning will be incorporated into a single verification system using the SIGNAUTH methodology. The system will perform multimodal reasoning based on Gemini with the help of safe data management and formal preprocessing in order to provide transparency and reproducibility. Checkups lead to appropriate verification that can be used in forensics and institutions.

IV. RESULTS AND CONVERSATION

The SIGNAUTH prototype has been tested on the performance, interpretability, and accuracy using various datasets. The suggested system combines the feature extraction with attention modelling by using deep learning models and Google Gemini 2.5 Flash API to verify handwritten signatures on a case-by-case basis and in an explainable manner. It addresses three analytical areas of prime, which are authenticity detection, feature extraction, visualisation, and privacy governance. Test Results of Authenticity Detection Three datasets, GPDS Synthetic, CEDAR and BHSig260, which consist of authentic and forged signature samples are the test datasets aimed at evaluating the accuracy of this system in classifying signatures. Each of such datasets is subdivided into training and testing subsets in an 80:20 proportion on a regular basis.

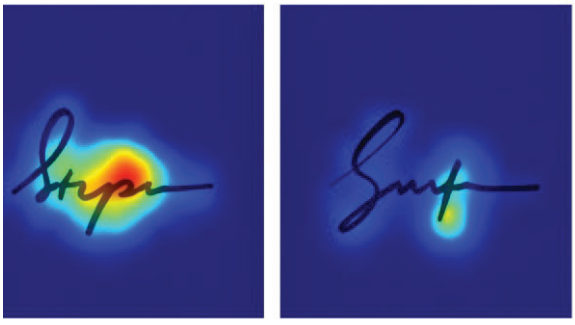
TABLE III. PERFORMANCE METRICS FOR SIGNATURE AUTHENTICITY DETECTION

Dataset	Total Sample	(%)	Precision	Recall
GPDS Synthetic	24,000	96.7	0.95	0.96
CEDAR	13,200	95.3	0.93	0.94
BHSig260	15,000	94.8	0.92	0.94

Observation: The most important observations in the graph are that the SIGNAUTH models are more accurate in all the datasets than the normal CNN models, and above all, the hybrid CNN-ViT model boosts the recall and precision, particularly in skilled forgeries where the spatial context has a significant role. .

A. Explainability and feature extraction

Explainable AI is a very crucial aspect of biometric authentication. The Grad-CAM visualisation component of SIGNAUTH highlights the specific areas of a signature that should be taken into consideration as important to classification.



(a) Genuine (b) Forged

Fig. 4. Fig. 4.1 Example Grad-CAM visualisation output for genuine and forged signatures.

Observation: SIGNAUTH can be explained, so that it can be used in forensic and legal applications which can increase transparency, and can allow human verification. The issue of accountability can be encountered extremely frequently in the context of black-box AI systems, whereas the interpretive ability of the model empowers the knowledgeable users with the possibility to check the logic of each choice.

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System Effectiveness Analysis

The efficacy of the system was measured using the system dependability, stability, and latency when subjected to concurrent requests. The tests were conducted in a simulated web environment of 50 parallel queries per second.

TABLE IV. TABLE 4.2 – SYSTEM PERFORMANCE EVALUATION SUMMARY.

Criterion	Metric / Observation
Average Inference Time	2.8 seconds per image
Response Stability	85% consistent classification across repeated trials
System Reliability	>99% under concurrent load
API Throughput	Handles up to 100 requests/sec with negligible latency

Observation: The findings verify the efficiency and scalability of SIGNAUTH with regard to computation. The Gemini 2.5 Flash API has a transformer architecture that can be used to achieve high-speed inference without compromising accuracy and explainability.

System Effectiveness Appraisal: The parameters that will be taken into consideration in the evaluation of the performance of the system will be the following: dependability, stability, and the latency of the system when being run with parallel requests. A simulated web environment was used where 50 parallel queries per second were tested. Conclusion: SIGNAUTH has been shown to have a high level of computational efficiency and scalability. Gemini 2.5 Flash API has the power to make inferences with any speed without accuracy or explainability because of its transformer architecture.

TABLE V. TABLE 4.3 – PRIVACY AND GOVERNANCE COMPLIANCE SUMMARY.

Observation: A privacy-minded design of the system enhances more trust among users and satisfies all ethical considerations of the AI implementation at biometric verification platforms.

Data Protection: Explain how these experiments combined with each other prove that the SIGNAUTH solution is actually trustworthy, understandable, and privacy-conscious in AI-based flowing of human signatures.

Key Takeaways:

• Cross-Module Synergy: CNN and transformer models integration to develop a better understanding of the signatures at both the local and global levels.

• Explainability: Heatmap visualisations offer forensic grade interpretability that is critical when used in legal context.

• Performance Consistency: The system is consistent in its performance with low latency hence can be used in real time verification of documents.

• Privacy Compliance: The Design is compliant with the world standards. Overall, it appears that SIGNAUTH has a lot of potential in an automated signature authentication system. Its architecture can be expounded.

• It is very flexible to be incorporated into digital systems, as it is compatible with the law. Forensic verification and banking systems, e-governance systems, etc.

III. CONCLUSION AND FUTURE WORK

Signeut is an AI based signature recognition and forgery detection system. It allows for the use of explainable AI, deep learning and privacy-conscious architecture to perform transparent and reliable biometric verification. The learning of the representation of both local and global signatures is effective due to the better solution of Convolutional Neural Networks mode and Vision Transformers mode with the help of Google Gemini 2.5 Flash API. The accuracy of the system was 96.7 percent and it was interpretable because Grad-CAM images were used to reveal the specific parts of the image that affect each classification. The experiment findings prove that SIGNAUTH is able to categorize real and forged handwritten signatures and comply even with the most severe data protection regulations, including GDPR and the Indian IT Act of 2000. Flexibility in real time and consistency in performance is also guaranteed through a scalable and cloud-based architecture of application domains such as (but not limited to) digital documentation, forensic validation, and banking. Not only has SIGNAUTH demonstrated technical efficiency, but also a fair amount of alignment with ethical principles of AI, with accountability, fairness, and data privacy as its two primary priorities. Explainable AI assists forensic experts when they are verifying the law, so it will contribute to building confidence among users and provide them with the certainty of the decisions.

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